

Midterm Lab Task 3 - Python List Collections

Problem

Midterm Lab Task 3 - Python List Collections

Problem 1. Using List Collection type. Create a program that will allow the user to perform the following **functions**: (add, update, search, delete, display, and sort) items in a list:

Note: You are free to decide what data you will be storing in the list and name the list based on the type of data you wish to store.

[MENU OPTIONS]

- 1 – Add Items
- 2 – Search for an Item
- 3 – Remove an Item
- 4 – View all items (Sorted either A-Z | Z -A)
- 0 – Exit program

Pick one [0 to quit]: ____

Requirements:

- The user can add items in the list until the user presses x to stop
- The user should be able to perform **search** if an item exists – Display if found or not found and count the number of instance in the list.
- The user should also be given the option to remove an item in the list – Display the Message “Item found and deleted” once deletion is performed – else display “item not found-deletion unsuccessful”
- The user may also opt to view items in the list and display items sorted in Ascending order
- The user may opt to exit the program by typing 0

Note: you are free to design the interface of the program, base on the Menu options shown.

Source Code

```

1 # Midterm Lab Task 3 - Python List Collections
2
3 def display_menu():
4     print("\n[ Skin Care Products ]")
5     print("1 - Add Items")
6     print("2 - Search for an Item")
7     print("3 - Remove an Item")
8     print("4 - View all items (Sorted either A-Z | Z-A)")
9     print("0 - Exit program")
10
11 def add_items(item_list):
12     while True:
13         item = input("Enter item to add (press 'x' to stop): ").strip()
14         if item.lower() == 'x':
15             break
16         item_list.append(item)
17         print(f"'{item}' added to the list.")
18
19 def search_item(item_list):
20     search = input("Enter item to search: ").strip()
21     count = item_list.count(search)
22     if count > 0:
23         print(f"Item found. Number of occurrences: {count}")
24     else:
25         print("Item not found.")
26
27 def remove_item(item_list):
28     item = input("Enter item to remove: ").strip()
29     if item in item_list:
30         item_list.remove(item)
31         print("Item found and deleted.")
32     else:
33         print("Item not found - deletion unsuccessful.")
34
35 def view_items(item_list):
36     if not item_list:
37         print("List is empty.")
38         return
39     order = input("Sort A-Z or Z-A? (Enter 'A' or 'Z'): ").strip().upper()
40     if order == 'A':
41         sorted_list = sorted(item_list)
42     elif order == 'Z':
43         sorted_list = sorted(item_list, reverse=True)

```

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43         sorted_list = sorted(item_list, reverse=True)
44     else:
45         print("Invalid choice. Showing unsorted list.")
46         sorted_list = item_list
47     print("Items in the list:")
48     for item in sorted_list:
49         print("-", item)
50
51 def main():
52     item_list = []
53     while True:
54         display_menu()
55         choice = input("Pick one [0 to quit]: ").strip()
56         if choice == '1':
57             add_items(item_list)
58         elif choice == '2':
59             search_item(item_list)
60         elif choice == '3':
61             remove_item(item_list)
62         elif choice == '4':
63             view_items(item_list)
64         elif choice == '0':
65             print("Exiting program. Goodbye!")
66             break
67         else:
68             print("Invalid option. Please choose again.")
69
70 if __name__ == "__main__":
71     main()
72

```

Output

```
[ Skin Care Products ]
1 - Add Items
2 - Search for an Item
3 - Remove an Item
4 - View all items (Sorted either A-Z | Z-A)
0 - Exit program
Pick one [0 to quit]: 1
Enter item to add (press 'x' to stop): Moisturizer
'Moisturizer' added to the list.
Enter item to add (press 'x' to stop): Toner
'Toner' added to the list.
Enter item to add (press 'x' to stop): Serums
'Serums' added to the list.
Enter item to add (press 'x' to stop): Masks
'Masks' added to the list.
Enter item to add (press 'x' to stop): Cleanser
'Cleanser' added to the list.
Enter item to add (press 'x' to stop): Sunscreen
'Sunscreen' added to the list.
Enter item to add (press 'x' to stop): x
```

```
[ Skin Care Products ]
1 - Add Items
2 - Search for an Item
3 - Remove an Item
4 - View all items (Sorted either A-Z | Z-A)
0 - Exit program
Pick one [0 to quit]: 2
Enter item to search: Serums
Item found. Number of occurrences: 1
```

```
[ Skin Care Products ]
1 - Add Items
2 - Search for an Item
3 - Remove an Item
4 - View all items (Sorted either A-Z | Z-A)
0 - Exit program
Pick one [0 to quit]: 3
Enter item to remove: Cleanser
Item found and deleted.
```

```
[ Skin Care Products ]
1 - Add Items
2 - Search for an Item
3 - Remove an Item
4 - View all items (Sorted either A-Z | Z-A)
0 - Exit program
Pick one [0 to quit]: 4
Sort A-Z or Z-A? (Enter 'A' or 'Z'): A
Items in the list:
- Masks
- Moisturizer
- Serums
- Sunscreen
- Toner
```

```
[ Skin Care Products ]
1 - Add Items
2 - Search for an Item
3 - Remove an Item
4 - View all items (Sorted either A-Z | Z-A)
0 - Exit program
Pick one [0 to quit]:
```